

Communities:

- Visually Impaired TTC Users in downtown - Muhammed Ali
 - Vision Loss Rehabilitation Canada
 - <https://visionlossrehab.ca/en>
- Disabled Powerlifters - Michael
 - [Parasport Ontario - Powerlifting](#)
 - info@parasportontario.ca
 - 416- 426-7187
- Elderly People who run into certain problems
 - [Sunshine Centres for Seniors](#) email sent, waiting for reply
 - [Home Page - Baycrest](#) email sent, waiting for reply
- Workers that shovel and remove snow
 - [Homepage - Snow Angels Canada](#)
- Masumin Basketball Association
 - <https://masumeenball.org/>
 - info@masumeenball.org

Meeting notes

Geographical area:

St. Catherine, Niagara falls, and Niagara on the Lake

Business model:

Fee for service no contract - university student

6-7 years

About 5-6 regular customers

Lawn mowing, snow shovelling

One client monthly contract

Techniques for clear driveway, lawn mow fastest

Advertising (paper to digital)

Isolated ideal customer profile, advertise in those neighbourhoods, reduce travel time

Agreeable clients, compatible

Established small network of subcontractors.

Difficulties

Refuse jobs for later due to snowstorm, subcontractors not available

Main issue: school schedule

Advertising (starting out)

Google business profile, google ad campaign, facebook ad campaign, door to door flyers and promotions

Communication issue: clinical shift, can't use phone (medical program)

New subcontractor, explain setup

Took on jobs too big, a lot longer, undercharging

Ideal client profile: older, retired, 50s, can't do it themselves, afford to pay the fee with haggling

Plan to go to medical school
Drop the price if scale, gas/car, not very profitable
25% profit margin
Not worth it?
Shovel and salt
Push mower and weed wacker, brooms and rakes
Snow mower and riding lawn mower (not for jobs), not worth it to buy a trailer
Equipment: old equipment break,
Lawn mower died on job, broke old lawn mower on job
Trouble removing ice, really thick
Shatter ice would be good

Team position

As engineering students who have taken a semester of Praxis, we have all deeply engaged in team work for various projects, such as the Praxis 1 design report and the CIV bridge project. We all heavily value the importance of communication in team work from our own experiences. In previous projects, there have been several occurrences of problems caused by insufficient communication or poor coordination of work distribution and scheduling. This time, we have decided to reduce this by breaking work into smaller parts, distributing responsibilities and setting deadlines ahead of time for each task on a task sheet. Keeping record of progress also greatly helps us stay aware of our pace.
(screenshot of task sheet)

The importance of clearance and efficiency in communication is not just internal, but also shown externally, especially with the communities that we get in touch with for the RFP. This is mainly shown in two different aspects. On one hand, when we draft emails, we try to keep them professional and simple at the same time so that the reader can accurately capture the core ideas that we want to express. On the other hand, we have also considered communities where there exist people whom our teammates know well so that early stage communication can be easier for both sides. For example, the community that we have chosen for the draft RFP is []. One of our team member's cousin has a personal business revolving around snow shovelling and lawn mowing, which provides us with great opportunity and convenience for interviews.

Besides communication, another thing we value is the relevance of the chosen topic to our own lives. Though this assignment doesn't specify the community must be close to us, we consider that choosing a topic that we are familiar with in our daily lives can give us solid foundational understandings of problems and makes it easier for us to identify opportunities and solutions. Most of our team members have lived in Canada for many years and deeply understand the trouble of removing snow in the winter. With a community that's closely related to our own lives, we have empathy and personal motivation to understand this community and we can draw on real experiences to test whether our interpretations make sense. Meanwhile, a potential risk caused by this is that we might over-generalize from our limited personal experiences and project our family's values or preferences onto the broader community. In order to prevent this,

we must keep our claims objective and get in touch with more than one individual from the community to obtain various perspectives.

As Praxis students, we are also going to apply the Praxis CTMFs we have learned in this course. Our Praxis I training provides structured approaches including Toulmin's argument structure, stakeholder analysis, requirements framework, diverging and converging, etc. These are all valuable systematic tools that

Research Regarding Optimization Algorithm

First Draft before we realised we had to not do solutions:

Purpose of This Draft

This draft document is a work-in-progress version of the team's final Request for Proposals (RFP). It documents our progress to date in understanding a community and exploring possible opportunities to better the lived experience of said community. It captures ongoing secondary research and primary interaction with the community. This document also serves as a support to a meaningful in-studio consultation and feedback from this will be used to refine the team's understanding and execution to aid the completion of our final RFP.

1. Team Position

As engineering students who have completed a semester of Praxis, we have gained extensive experience working on projects such as the Praxis I design project and CIV bridge project. Through these experiences, we have come to place strong value on clear communication and coordination, due to previously encountering challenges arising from misaligned expectations and uneven task distribution and poor time management. Our current approach involves emphasizing dividing work into smaller clearly defined tasks, assigning responsibilities with internal deadlines, and maintaining constant updates on our progress. Our task sheet, as seen in **Figure 1**, shows how as a team we have approached organization in our team.

Task	Assigned to	Due by	Done	Notes
Finish preliminary contact with communities	Everyone	Jan 22, 2026 (Thursday)	☒	
Contact MBA and Cousin for lawn mowing/snow plowing	Hasnain	Jan 22, 2026 (Thursday)	☒	
elderly's home (backup)	Frank	Jan 22, 2026 (Thursday)	☒	
Schedule Meeting with cousin for lawn mowing/snow plowing	Hasnain	Jan 26, 2026 (Tuesday)	☒	
Reply to Vision Loss Rehabilitation Canada	Muhammed Ali	Jan 27, 2026 (Tuesday)	☒	
Develop Team Position	Everyone	Jan 27, 2026 (Wednesday)	☒	
Finalize Community	Everyone	Jan 27, 2026 (Wednesday)	☒	Decided on snow shoveling community
Draft RFP Final Copy	Everyone	Jan 31, 2026 (Saturday)	☐	
Finalized Team Position	Frank	Jan 30, 2026 (Friday)	☒	
Research for all the opportunities	Everyone	Jan 31, 2026 (Saturday)	☐	
Figure out Opportunity 3	Everyone	Jan 30, 2026 (Friday)	☐	
Convert all secondary research into paragraph form	Hasnain	Jan 31, 2026 (Saturday)	☐	
Stakeholders		Jan 31, 2026 (Saturday)	☐	
Lived Experience	Hasnain	Jan 31, 2026 (Saturday)	☐	
Contextual Factors		Jan 31, 2026 (Saturday)	☐	
Opportunity 1 Research	Michael	Jan 31, 2026 (Saturday)	☐	
			☐	

Figure 1: Task sheet with members and deadlines labelled and updated constantly

The importance of clearance and efficiency in communication is not just internal, but also shown externally, especially with the communities that we get in touch with for the RFP. This is mainly shown in two different aspects. On one hand, when we draft emails, we try

to keep them professional and simple at the same time so that the reader can accurately capture the core ideas that we want to express. On the other hand, we have also considered communities where there exist people whom our teammates know well so that early stage communication can be easier for both sides. For example, the community that we have chosen for the draft RFP is independent snow removal contractors. In this case, one of our team members has a personal connection with the business owner.

In addition to communication, we value the relevance of the chosen topic to our own lives. Choosing a topic that we are familiar with in our daily lives can give us solid foundational understandings of problems and makes it easier for us to identify opportunities and solutions. Most of our team members have lived in Canada for many years and deeply understand the trouble of removing snow in the winter. With a community that's closely related to our own lives, we have empathy and personal motivation to understand this community and we can draw on real experiences to test whether our interpretations make sense. Meanwhile, a potential risk caused by this is that we might over-generalize from our limited personal experiences and project our family's values or preferences onto the broader community. In order to prevent this, we must keep our claims objective and get in touch with more than one individual from the community to obtain various perspectives. Any perspective that we put into this as a claim, must have proper justification and evidence to minimize personal bias as much as possible and create something that would in turn benefit the broader community, not just those members that share our values.

As Praxis students, we are also going to apply the Praxis CTMFs we have learned in this course. Our Praxis I training provides structured approaches including Toulmin's argument structure, stakeholder analysis, requirements framework, diverging and converging, etc. These are all valuable systematic tools that can inspire meaningful engineering thinking and discussion throughout this project. However, we will definitely not be constrained by certain structures or concepts, but integrate these with what we are learning in Praxis II and our real-life experiences with engineering designs.

2. Community Overview

2.1. Community Definition

The target community consists of independent, small-scale snow removal business owners serving residential clients in the GTA as their shared interest. Zain Ahmed is one of these business owners and is our main point of contact and representative of this community. There is a conservative estimate of 300-500 individual or small snow removal contractors in the GTA, judging from active advertisement posts on facebook

and kijiji. Here are some key aspects or common interests that members of this community shares:

- Business entrepreneurship: They are responsible for all aspects of their business, including client acquisition through local paper or digital advertising, developing and adjusting pricing strategies, and personally investing in and maintaining their own equipment
- Operational challenges: They face similar challenges in winter in this area, which includes difficulty of travel in snowstorms, equipment breakdowns, need for manual work in freezing temperatures, and the unpredictability of income due to varying weather conditions.
- Client relationships: Instead of having a corporate network like larger companies, these individual business owners acquire new clients mainly by advertising locally, by paper or digitally. They contact existing customers directly and tend to develop stable and long-term loyal customers.
- Seasonal Nature: The operation time of the business only lasts 4-5 months. Therefore, they might have other businesses (such as lawn mowing) in the summer or additional income sources.
- Physical demand and safety: Members of this community usually have to do manual labour in extreme cold, leading to risk of frostbite or exhaustion. There are also risks of injury from slip or equipment operation or repetitive strain.

2.2. Stakeholders for this Project

2.2.1. Zain Ahmed: Snow Removal Business Owner (Primary Stakeholder)

As a business owner, his interest is to secure clients and maximize revenue, which involves providing high quality services. However, safety and efficiency are equally as important for both his clients and his sub-contractors.

2.2.2. Individual Snow Removal Business Owners in the GTA (Secondary Stakeholder)

Direct members of the target community. Their core interests are to secure clients and maximize revenue, which involve providing service of quality. Safety and efficiency of work are important considerations. Building and maintaining reputation among clients are also crucial.

2.2.3. Snow Removal Services Clients (Secondary Stakeholder)

Their main interest is to have a safe and clear driveway/walkway in front of their houses during snow season. They are looking for reliable service and reasonable pricing. Minimizing property damage is one important factor when they are choosing services.

2.2.4. Equipment retailers and Manufacturers (Tertiary Stakeholder)

Modifications to existing equipment or new equipment would have influences on their production process if applied. They aim to make profit by selling equipment and maintenance services to contractors. Quality, efficiency, and cost of equipment and tools are all significant factors taken into consideration for manufacturers.

2.2.5. Design Team (Primary Stakeholder)

The four team members in this Praxis II design team are also stakeholders for this project. As first year engineering science students, this project can help us gain more practice in engineering design, applying learned CTMFs and have a deeper understanding of engineering that we can bring into any future work.

3. Understanding the Community

In an online meeting with a member from the target community, we interviewed him around topics about his business routine, equipment operation, advertising methods, clients, difficulties and challenges in order to identify potential opportunities for improvement where engineering design can come into play. Talking to a member of the community has given us a deeper understanding of the community itself. Here are some inspiring aspects that the interviewee mentioned that influenced our diverging process. The full interview transcript can be found in Appendix A:

- He has transformed from advertising on paper at the beginning to mainly advertising digitally on platforms now.
- He has about 5-6 regular customers. Most of his customers are elderly who aren't physically suitable for shovelling snow by themselves and can afford snow removal services.
- As a university student who has to balance studies with this job and scheduling has been a problem for him during busy seasons.
- He has established a small network of subcontractors. Some of them can work in place of him whenever he is not free.
- Explaining setup and related knowledge required to new subcontractors can be complicated.
- He explained the difficulty of removing ice that sticks to the ground and claimed that it would be great if something can shatter the ice for him.
- He isn't planning to scale up his business because competition in the market would force him to lower his pricing and that he would have to pay more subcontractors. It is considered overall not worth it.
- Broken old equipment on site during work can be a real pain.

These lived experiences show that there are needs in various aspects of this job, which means that there is plenty of room for designs for improvement.

4. Candidate Engineering Opportunities

4.1. Opportunity 1: Training New Sub-Contractors

As per the interview with Zain Ahmed [See Appendix A], he says “

In order to maintain efficiency of snow shovelling, different methods of path taking can be measured to determine a more optimal split. Improvements in business output can be obtained by developing “techniques to clear a driveway the fastest”, as stated by Zain Ahmed. This opportunity aims to explore developing an algorithm to quantitatively measure how efficient someone was at shovelling snow. Additionally, we wish to employ some other features which can graph out the acceleration and angle of attack.

Following the main idea of this opportunity, we wish to store data values for the location, force generated for each time someone shovels, angle of attack, and time taken. The main hardware stack, as a broad overview, will contain a microcontroller acting as a main control system, a GPS module, an inertial measurement unit measuring acceleration, and finally a SD card module for data storage. Noisy raw data taken from the shovelling experiment can then be filtered out. After all is complete, software to track the data and determine a score when compared to an optimal pathway can be calculated.

Major stakeholders would contain newer subcontractors who might not be as experienced in shovelling efficiently. These physical output systems could help improve their performance, resulting in an opportunity to save more time.

4.2. Opportunity 2: Ice Removal from Driveways

This opportunity was identified from the discussion we had with the representative of our chosen community (transcript included in [Appendix A](#)). It identifies the problem of ice that has been strongly bonded to the pavement. Typical shovelling is not effective and this method requires great strain on the user to remove the ice. The representative indicated that currently he simply salts it and lets it be, hence potential solutions for this issue would aim to completely remove the ice as an improvement on the already existing method that is status quo.

4.2.1. First approach: Targeted Thermal Weakening

This approach would make use of a heating to melt the thin adhesive layer that exists on the ice-pavement boundary. battery powered sharp metal heating element that can break adhesive bonds in the adhesive layer between the pavement and the ice. The energy used to weaken the layer is similar to that of a

household microwave (see [Section 6.1.2](#)) and can be easily achieved from a battery.

4.2.2. Second approach: Vibration/Repeated Impact Based Bond Weakening

This approach makes use of a motor driven mechanism that deals repeated blows to the adhesive layer. It slowly chips away at that layer until the ice can be separated from the pavement with relative ease. It will require some form of power supply to function.

5.3 Opportunity 3: Job Scheduling + Quoting Software

One of the major business considerations would be determining how much to quote each customer based on square feet of the areas that require shovelling. Additionally, there is a more hidden cost of gas money required to get to a certain location. Taking these two things into consideration, it allows for contractors to get a better idea on how lucrative a certain job opportunity could be.

Our third opportunity entails a software based solution where potential customers are able to file their location along with the area of their driveways and sidewalks. By determining how far the location is for a contractor, they are able to determine how expensive it would cost for gas money to get to the job. Additionally, the area of the driveway allows for the worker to get a good sense on what to quote the customer. These two factors would allow for a comprehensive outlook for the contractor to make a smart business decision.

5. Research Constructed to Date

5.1. Opportunity 1: Shovelling Path Optimization for New Contractors

- Tracks the person shovelling and their path for clearing a driveway while mapping the entire area
- Use an algorithm to take the input data and determine the inefficiencies such as an inefficient shoveling path taken or wasted time doing nothing
- Generate and teach the new operator on more efficient shoveling that otherwise would only be obtained by years of experience

Related Solutions:

- Geotab telematics is related
- Data-driven insights to drive operational efficiency of large fleets of trucks
- Has real-time vehicle tracking and stores data on driving behaviour

- Has hardware which tracks the location of the truck, acceleration, and any sharp turns, determining efficiency of the drive

5.2. Opportunity 2: Ice Removal from Driveways

5.2.1. First approach: Targeted Thermal Weakening

- **Core Principal**
 - Ice sticks to pavement through mechanical interlocking (when ice goes into the cracks of the pavement and fills surface pores) and adhesive bonds. Fully breaking or melting the ice is likely to take too much time and energy. Hence, for this opportunity it is best to find an approach that gets rid of the adhesive layer.
 - Reference to Adhesive Bonds: [\[1\]](#)
- **First approach: Targeted Thermal Weakening**
 - Localized heating at the ice-pavement boundary, breaking adhesion, does not fully melt the ice.
 - Can be delivered via heated metal element.
 - More realistic and less power needed as we only need enough heat to weaken adhesive bonds, not fully melt the ice. The energy used to weaken ice adhesion is similar to a household microwave, so it is reasonable to get from a tool which charges or uses a battery.
 - See calculation in [Appendix B](#).
 - Key challenges:
 - The space needed to store the battery on the tool itself, could increase the weight by a lot, making it heavy and inconvenient.
 - Safety risk exists with heating elements.
 - Ensuring the electrical parts of the tool do not go bad in extreme cold.
 - Ensuring the heating element causes no damage to pavement.
- **Second approach: Vibration/Repeated Impact Based Bond Weakening**
 - Spring Loaded and Motor Driven Mechanism that repeatedly chips at the adhesive layer.
 - Like a scaled down jackhammer.
 - Due to the brittle nature of ice, this method is likely to work.
 - Requires less sustained user force.
 - Key challenges:
 - Limiting damage to concrete & asphalt.
 - Distributing load over a large impact area.

- Tool Stability.

5.3. Opportunity 3: Job Scheduling + Quoting Software

- A software that can easily allow users to book shovelling and input key details such as the location of their home and size of the driveway, automatically calculating and quoting prices, will automate a manual process the snow shovelling business owner currently does, reducing friction and effort.
- The reduced friction and availability of a website on which the software exists would boost sales for a specific product.
- It is widely documented in digital marketing that the existence of ‘Landing Pages’ and websites for the user to confirm their order themselves boosts higher sales conversion rates and can be optimized with principals [7].
- Additionally, price transparency at the time of the checkout increases how many customers actually convert [8].

6. Next Steps

This draft represents our team's progress through secondary research and preliminary analysis. Moving on, here are some steps that we will take towards the final outcome of the project:

- Refine and validate each opportunity and further explore the solution space.
- Continue reaching out to the community and discuss the opportunities and potential solutions with them to validate our direction.
- Conduct more in-depth secondary research on the design opportunities that we have identified to generate early-stage prototyping ideas for designs.
- Gain insight of feasibility of potential solutions
- Do more iterations of converging and diverging to more clearly define our current options.
- Try to hold interviews with other small-scale snow removal contractors and some of their clients.

7. References List

[1]

H.-C. Dan, L.-H. He, J.-F. Zou, L.-H. Zhao, and S.-Y. Bai, “Laboratory study on the adhesive properties of ice to the asphalt pavement of highway,” *Cold Regions Science and Technology*, vol. 104–105, pp. 7–13, May 2014, doi: <https://doi.org/10.1016/j.coldregions.2014.04.002>.

[2]

H. Chen, Y. Wu, H. Xia, B. Jing, and Q. Zhang, “Review of ice-pavement adhesion study and development of hydrophobic surface in pavement deicing,” *Journal of Traffic and Transportation Engineering (English Edition)*, vol. 5, no. 3, pp. 224–238, Jun. 2018, doi: <https://doi.org/10.1016/j.jtte.2018.03.002>.

[3]

Bosch Sensortec, “BNO055 Intelligent 9-axis absolute orientation sensor,” *Digi-Key*, Oct. 2021. <https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bno055-ds000.pdf> (accessed Jan. 30, 2026).

[4]

Espressif, “Espressif Documentation,” *Espressif.com*, 2025. https://documentation.espressif.com/esp32_datasheet_en.pdf

[5]

Quectel, “Quectel LC29H Series Dual-Band Multi-Constellation GNSS Module with RTK and DR Functions,” Quectel, Jun. 2022. Accessed: Jan. 30, 2026. [Online]. Available: https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/7133/Quectel_LC29H_Series_GNSS_Specification_V1.8.pdf

[6]

Kingston Technology, “microSDHC memory card Flash Storage Media,” Kingston Technology, Oct. 2019. Available: https://www.kingston.com/datasheets/SDCIT-specsheet-8gb-32gb_en.pdf

[7]

H. Meslem and A. Abbaci, “A practical approach for optimizing the conversion rate,” *Marketing Science & Inspirations*, Dec. 31, 2022. <https://msijournal.com/practical-approach-optimizing-conversion-rate> (accessed Jan. 01, 2026).

[8]

L. Xia, K. B. Monroe, J. L. Cox, and J. M. Jones, “The Price Is Unfair! A Conceptual Framework of Price Fairness Perceptions,” *ResearchGate*, Nov. 2004. https://www.researchgate.net/publication/228590264_The_Price_Is_Unfair_A_Conceptual_Framework_of_Price_Fairness_Perceptions (accessed Jan. 30, 2026).

8. Appendices

8.1. Appendix A: Meeting Transcript with Zain Ahmed for Snowshoveling

Hasnain Heryani: 15:25

Could you explain in depth what your business model is?

Zain Ahmed: 15:51

Are you asking about the financials?

Hasnain Heryani: 15:55

So I'm asking about, I guess if that's the best way for you to explain it, then go ahead. You can start with that and then have some follow up questions.

Muhammed Ali Bhanji: 16:08

I think more so like also how, like how many people you service and how often you do it, you know, so that kind of stuff as well.

Zain Ahmed: 16:14

Okay, I see. So it's fee for service. I don't do like any contracts or anything like that just because I'm also a university student. So I've lost count of how many customers I've had over the years because I've been running the business for, I want to say about six years, six or seven years. And I have, I'd say about five regulars. Yeah, about five or six regulars. And then whenever they need me, they call me out for lawn mowing or snow shoveling. And that's basically it. During the summer I do have a contract with one client where they pay monthly and that's it. The rest are fee for service.

Hasnain Heryani: 17:06

Okay, I see. So the answer as well. So could you then also explain over the six or so years that you have been running your business, what are some steps that you've taken to make it more efficient?

Zain Ahmed: 17:29

To make it more efficient? Well, I would say one I've learned just like the techniques to clear a driveway and mow a lawn the fastest. I also automated advertising for a while and I started moving to digital advertising over paper advertising. Another thing I did was once I. I don't want to use too much like business lingo, but once I kind of isolated my ideal customer profile, I started to kind of advertise in those neighborhoods to minimize the travel time between jobs as well as to just kind of make the clients I have agreeable and more compatible with my business model. And then also I've established kind of a small network of, I want to say, subcontractors. So sometimes like if I'm gone, say on vacation, I can call on one of them and chances are they'll take the job and do it for me while I'm gone.

Hasnain Heryani: 18:53

Okay, so then also as a university student yourself, do you find that there are any logistical or communication difficulties that you find for finding people to cover for you when you're traveling or for. With those contractors that you spoke about?

Zain Ahmed: 19:23

Yes, there are logistical issues. Sometimes, like, I straight up have to refuse a job or reschedule for later. If there's like a snowstorm on a day where I'm in clinical or I'm in classes all day, then like, a lot of the times I just can't make it out till the day after the snowstorm. And sometimes, like, I reach out to my subcontractors and they're also unavailable because of their school schedules. So, yeah, there definitely are logistical challenges sometimes.

Hasnain Heryani: 19:58

Okay, so would you say that besides the school schedule, there are any other causes to these logistical issues.

Zain Ahmed: 20:11

Right now? I would say no. That's no major ones. The school schedule is the main logistical issue.

Muhammed Ali Bhanji: 20:22

And so I have a question about this as well, I guess. Okay. School stuff is like the main logistical issues. Right. But what about, like, smaller quality of life issues? Because for us, what we're trying to do is like, service the community, which is like people like you who have lawn mowing businesses, including you, lawn mowing and snow shop, like that. Businesses like you. And we want to improve any sort of aspect of it. Right. So is there any, like, kind of quality of life, smaller things that could kind of become easier, like problems you face?

Zain Ahmed: 20:53

It'll make it easier. Honestly, I'd say with my operations, no right now, but maybe with advertising, like, adjusting to the advertising was like, kind of difficult. So especially starting out. So maybe if I had like, more support or like an easier platform to advertise digitally when I started out, that would have helped maybe.

Ziqi Yin: 21:25

So could you go a little deeper in how you advertise right now?

Zain Ahmed: 21:30

So right now I'm not advertising as much just because I'm also studying for the mcats. So I kind of had to put the business on pause. But when I was advertising at the peak, I basically had a Google, my business profile, and I still have it. And I was basically running Google Ad campaigns. I also ran a Facebook ad campaign, but it didn't do as well. So I focused more on Google Ads and then door to door, like flyers and sales.

Ziqi Yin: 22:03

Thank you.

Hasnain Heryani: 22:04

I see. All right, So that's in terms of logistical. How about on the communication side of things? So you said you have many subcontractors, those being your friends that cover for you when you're out of town or you're unavailable for a specific job. Do you find that when you are looking for any of these friends that may be available to help you do you have any communication issues? Are you looking to, like, repeatedly explain things and are there issues there, perhaps Any difficulties you find?

Zain Ahmed: 22:49

Not really. The only communication issue sometimes is if, you know, if I'm say it like a clinical shift, like, if I'm really busy, then it can be kind of difficult. If I can't use my phone while I'm basically at, like, this school event, it's called, like, a clinical shift. It's part of my program. And then sometimes, like, if it's a new subcontractor, then sometimes, like, having to explain, like, the equipment setup and how that works can be a little difficult, especially if they have questions. But usually it's not too bad.

Ziqi Yin: 23:26

So in the process of serving clients, what is. Could you remember any encounters where it was. It was really hard to encounter problems that you couldn't deal with, especially when you're doing the job?

Zain Ahmed: 23:45

There have been in the past, especially when the business was new, where I took on jobs that were, like, too big, that, like, I just couldn't do, and it ended up taking a lot longer than I thought it would, and I ended up undercharging. But other than that, I can't think of any.

Hasnain Heryani: 24:09

So could you define the client profile that you talked about, what you found to be your ideal client profile?

Zain Ahmed: 24:18

So usually I'm looking for someone who is older, usually retired, or kind of maybe not retired, but usually in their 50s. So, you know, people basically who kind of can't do it themselves, or it's difficult for them to do it by themselves, but they're still affluent enough that they can afford to pay the fee without, you know, without basically, like, haggling. Right.

Hasnain Heryani: 24:46

Okay.

Michael Lin: 24:49

Do you see, like, the snow shoveling business and lawn mowing business is something that, like, you can go on a larger scale because I know you're mainly, like, you're not exactly a very big team right now, but do you see, like, someone potentially making this, like, a.

Zain Ahmed: 25:02

Very big business for me right now just because I plan to go to medical school after I graduate? No, theoretically it could be if, say, I wasn't interested in medical school, it could definitely be scaled. The only issue is, is that if, say, I charge, like, if, say I scale, I'll have to drop the average price from, say, like 40 to \$50 per job to around, like \$40. And then if I pay an employee, I'll basically have to pay them probably around \$25 for their labor and the gas for their car. And then that basically leaves about 10, \$15. But then I'll also have to get like insurance and like there'll be equipment maintenance and stuff like that. So all in all I'll probably be left with around \$10 profit per job as like a gross operating margin, but then like in gross profit. But the issue is that then only leaves me about \$10 to advertise per, per potential client. So it could like theoretically work. 25 gross profit is like good. But after I like once all is said and done and I factor in advertising costs, my profit margin would probably drop to about 10%. Which for big businesses and just medium sized businesses in general, that is actually an average net profit margin. But I feel like if I was going to put in that much time and energy and money, I'd rather do it into something with more ROI.

Hasnain Heryani: 26:48

Okay, so could you then next explain the equipment that you use and specifically what kind of equipment you're using to make your life easier in getting these jobs done.

Zain Ahmed: 27:05

So right now it's pretty basic. Like I just use for snow shoveling I just use a shovel and salt. And then for lawn mowing I use a push mower and like a, I have a weed whacker, like an edger and then obviously just like brooms and rakes and basic stuff like that. I do actually for my house, for personal reasons, have a snowblower and a riding lawnmower. But the only issue is, is that to use those for jobs I would have to buy a trailer. And right now just because again, because of my school schedule and MCAT studying, it doesn't make sense for the financial investment into a trailer. So that's the only reason I don't do that.

Ziqi Yin: 27:51

And in the process of operating the tools that you have mentioned, are there problems that you might encounter in certain situations? Like for example, if, if the snow is really. There's a lot of snow, if there's a snowstorm or anything like that.

Zain Ahmed: 28:06

Yes.

Ziqi Yin: 28:06

Would it be more difficult for it to operate those machines? And could you explain it a little bit?

Zain Ahmed: 28:12

So sorry, you're asking about like equipment like failure and complications. Is that what you're asking?

Ziqi Yin: 28:19

Not necessarily like specifically that, but could include that basic, generally any kind of problem when operating.

Zain Ahmed: 28:27

So usually. So there are definitely problems. I have broken like rakes and shovels on jobs.

Unfortunately, especially when starting out, I was using a lot of like old equipment that we just kind of had lying around. I did also break, actually I had an old lawnmower and I did have an issue once where I think it died on a job and it needed like an oil change, which was kind of stressful. And I kind of had to call my dad and like, troubleshoot. I did also break my old lawnmower on a job, so I actually had to. I was almost finished the job, thankfully, so I was able to use just my weed whacker to clean it up and finish. And then I had to get a new lawnmower. But that's really it. Just equipment breaking and like, there being issues that need to be troubleshooted. That's it.

Muhammed Ali Bhanji: 29:23

You said that you use a shovel, right? Mainly on jobs.

Zain Ahmed: 29:26

Yeah.

Muhammed Ali Bhanji: 29:27

So, yeah, sorry. So when you have that, do you have any problems ever, like, kind of shoveling certain parts where you either feel like you can't get to certain parts or it could be more efficient. Like, for example, when I shovel, I have, like, a problem getting, like, under the railing of the stairs, that kind of stuff. So anything like that?

Zain Ahmed: 29:44

There are some times where, like, I have trouble removing ice if it's just like, very thick, Especially at the snow bank at, like, the front of the driveways. But other than that, not really. Because even like the area. Usually when someone hires you to shovel a driveway, they're less

interested in it kind of like looking pretty, and they're more interested in them being able to get their cars in and out.

Muhammed Ali Bhanji: 30:09

Yeah, that's fair.

Hasnain Heryani: 30:17

Do any. Do any of the rest of you guys have any other questions? No.

Muhammed Ali Bhanji: 30:26

One thing about the. The ice, Right. Just follow up on that. You said the main thing is that the ice kind of gets stuck and then that's harder to travel. Of course. Right?

Zain Ahmed: 30:33

Yeah.

Muhammed Ali Bhanji: 30:34

If. Okay, just thinking aloud here. There's hypothetically something that kind of.

Hasnain Heryani: 30:41

Split.

Muhammed Ali Bhanji: 30:41

It up into smaller pieces and you could travel that. Would that kind of make your life easier with that specific problem?

Zain Ahmed: 30:47

If. What?

Muhammed Ali Bhanji: 30:47

Sorry, that's just thinking aloud here. But let's say there was something that kind of like shattered the ice for you, right? Instead of you having to do it. That would help.

Zain Ahmed: 30:56

Yeah, yeah, that would help. Right now I just. If I encounter ice like that, usually, like, I just clear as much as I can throw down salt and call it a day. So, yeah, having something like that would definitely help.

Muhammed Ali Bhanji: 31:10

Okay.

Hasnain Heryani: 31:16

So, Zane, so last thing I think, before we end off the meeting is that if we were looking to maybe do a short interview with any of your clients, would that be a possibility or.

Zain Ahmed: 31:31

No, unfortunately not. Just because. Just because, like, a lot of them are older and also I don't meet a lot of them face to face. A lot of them either are older and, like, are Very technically incapable. Like, they can't even use, like, E transfer. A lot of time they have to pay, you know, in cash. And the ones that are usually younger and able to pay are either done through intermediaries, like real estate agents, or they're, like, you know, overseas. And they're just hiring me remotely so I can ask, but most likely not, unfortunately.

Hasnain Heryani: 32:13

All right. Okay. Thank you for your time.

Zain Ahmed: 32:17

No problem.

Ziqi Yin: 32:17

Thank you very much.

Hasnain Heryani: 32:18

I appreciate it.

Zain Ahmed: 32:20

Hope it helped.

Muhammed Ali Bhanji: 32:23

Yeah, thank you so much. Yeah, thank you.

Hasnain Heryani: 32:26

Have a good day.

Zain Ahmed: 32:27

Thanks, you too. Take care. Good luck.

Hasnain Heryani: 32:29

Thanks.

8.2. Appendix B: Calculations for Targeted Thermal Weakening

Assumptions

Ice temperature: $-10\text{ }^{\circ}\text{C}$

We heat the bottom layer to $0\text{ }^{\circ}\text{C}$ and melt a thin layer of thickness t

Ice density: $\rho = 917\text{ kg/m}^3$

Specific heat of ice: $c = 2100\text{ J/(kg}\cdot\text{K)}$

Latent heat of fusion: $L_f = 334,000\text{ J/kg}$

Area basis: $A = 1\text{ m}^2$

Formulas

Mass of the interfacial layer: $m = \rho A t$

Energy to warm ice from $-10\text{ }^{\circ}\text{C}$ to $0\text{ }^{\circ}\text{C}$: $Q_{\text{warm}} = mc\Delta T$

Energy to melt that layer: $Q_{\text{melt}} = mL_f$

Total: $Q_{\text{total}} = Q_{\text{warm}} + Q_{\text{melt}}$

Calculations

If you melt 0.5 mm at the interface

Mass: $m = 0.4585\text{ kg/m}^2$

Warm $\approx 9.6\text{ kJ/m}^2$

Melt $\approx 153.1\text{ kJ/m}^2$

Total: $\approx 162.8\text{ kJ/m}^2$